CSC 471 - Mobile Application Development (iOS)

Final Project Proposal

Project Title - FactoHealth (*Health information provider*)
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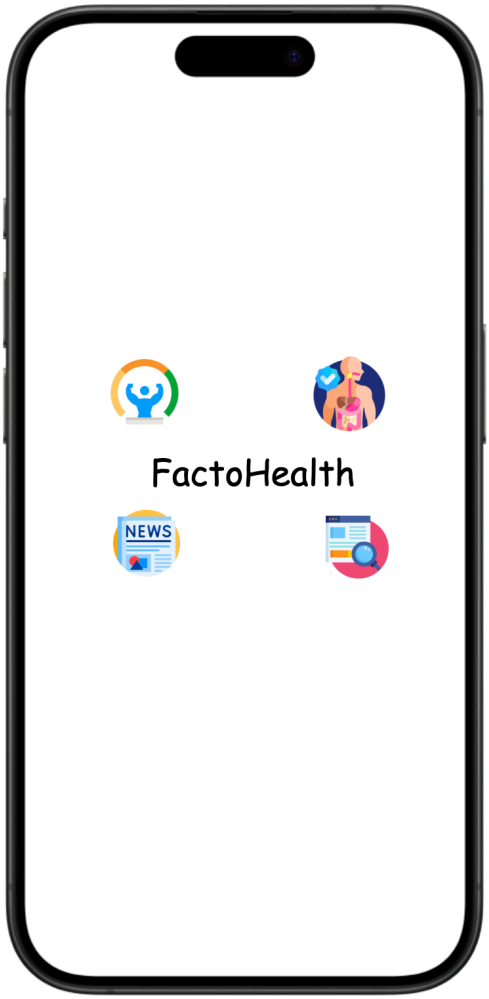
This is an **Individual** Project

Project Description - The app is imagined to be a one-stop shop for any health related information. With rise in health consciousness, the app will help provide a deeper look into the latest research in the health industry being undertaken by some of the worlds leading Pharma techs and institutions. To address the curiosity of self health, the app provides a BMI calculator along with a series of "know your organs" descriptions and lastly a news feed with the latest happenings in the health industry. A unique feature of the app is that it will scan the medicine picture and display the medicine details.

Preliminary screen design -

Screen 1 - Launch Screen

1

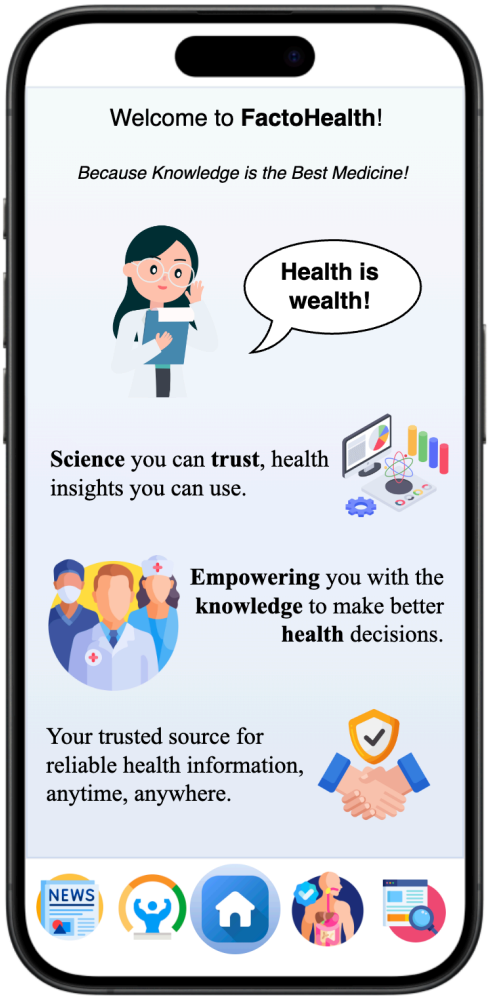


A simple Launch Screen that appears before the app loads with a splash screen implementation to animate the movement of the logo. The four icons move in from different directions and the text fades in before transitioning to the main screen.

This is a marquee message

Screen 2 - Main Screen

2



Main screen being considered as a landing page welcomes the users and provides a navigation bar at the bottom of the screen.

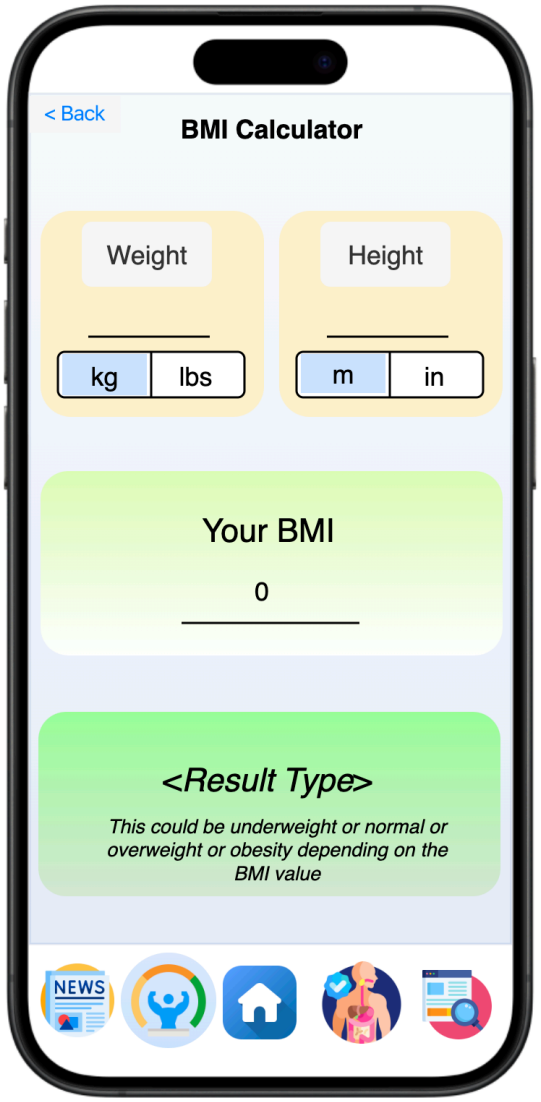
Screen 3 - News Feed Screen

3

This will be a scrollable view with similar stack views for a list of news articles as per available from the API or hard coded links

Screen 4 - BMI Calculator Screen

4



The "News Feed screen" displays news specifically filtered for the Health category of the news feed.

The news slider will have top headlines for the health category.

It will be a scrollable view, ordered from the latest to the oldest.

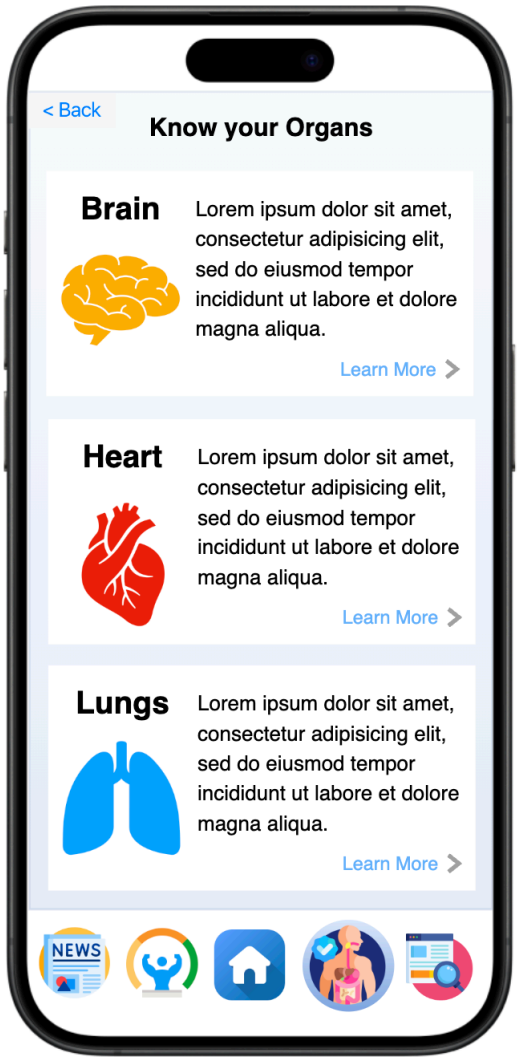
Certain News APIs such as "**NewsAPI**" or "**Google News API**" or "**Mediastack**" will be used to cater this newsfeed. These APIs provide at least 100 free requests per month which will be sufficient for the project implementation.

The "BMI Calculator Screen" is designed to perform standard **BMI** (*Body Mass Index*) calculations and display results as per standard ranges of BMI value based on weight and height.

Another factor which has been followed in many such calculators found online and in this implementation is the flexibility to input values for weight and height in both **Metric** and **Imperial measurement systems** to cater to wider audience without the need to convert values to either of the systems.

Screen 5 - Know Your Organs Screen

5

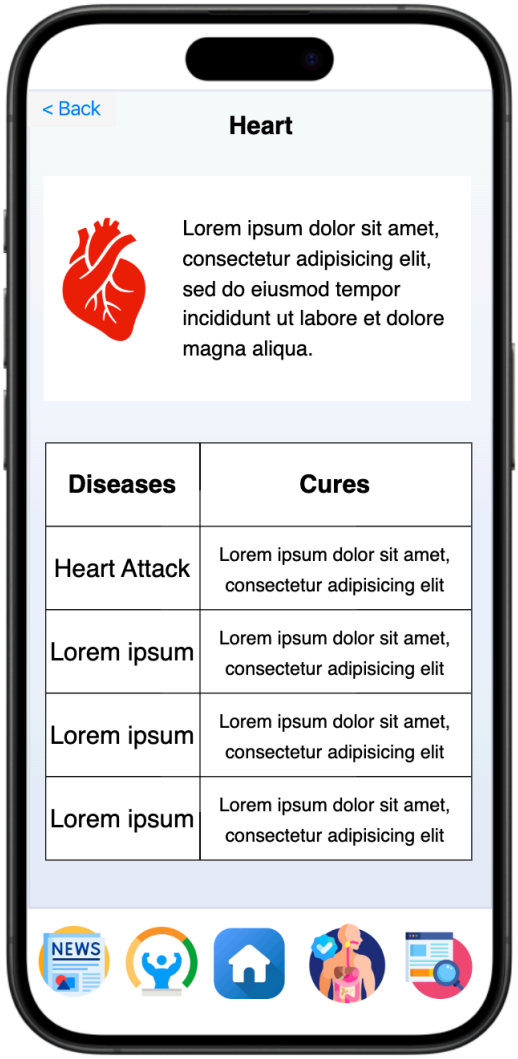


When the " Learn More " button is clicked it will redirect to respective organ sub-screens as shown here

This will be a scrollable view with similar stack views for Brain, Heart, Lungs, Pancreas, Kidney and Intestines

Screen 5a, 5b, … ,5f - Sub-Screens

5a, 5b, … ,5f



"Know Your Organs" screen displays a list of organs with respective diagrams and summary.

This screen is also a scrollable for limited organ summaries.

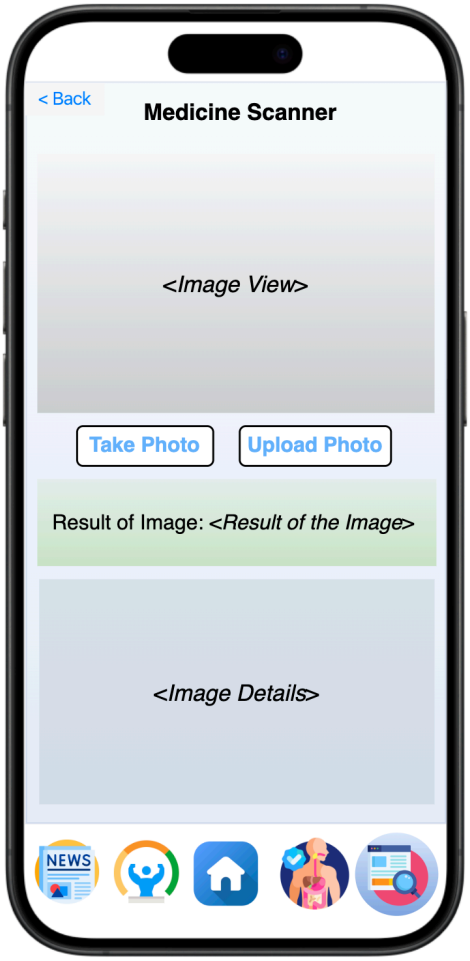
Each of the organ summary further leads to sub-views with the tap of the "**Learn More**" present to the bottom right corner.

This sub-view screen will provide the complete details of the organ, including the diseases and cures in a simple table.

NOTE: All the data displayed here will be hardcoded data and not be fetched using any APIs.

Screen 6 - Medicine Scanner Screen

6



"Medicine Scanner" screen is designed to provide users with the medicine details based on the image of the medicine.

Here, users are given a choice to **take photo** of the medicine via the rear camera or **upload photo**.

Using Image recognition from Core ML API, details of the medicine will be provided in the Result.

Details such as Medicine name, purpose, Dosage, warnings etc will be provided to users to have a better understanding of the medicine that is being consumed by just taking a photo or uploading the photo.

NOTE: The model will be specifically trained only for medicines and APIs like OpenFDA API or RxNorm API will be used to provide the Medicine details.

API features used -

The app uses the below API features,

1. **Quartz 2D drawing** - For 2D diagrams of the organs and few other animations wherever necessary.
2. **Multi-touch events** - Such events may involve "zoom" and other events wherever necessary.
3. **CoreML** - Mainly used in the "***Screen 6*** - *Medicine Scanner*" for performing the operations necessary to provide medicine details via processing the image.
4. **News API** - Certain news API's like "**NewsAPI**", "**Mediastack**", "**Google News API**" will be used for filtered news related to health industry.
5. **OpenFDA API** or **RxNorm API** - Will be used to fetch additional details about the medicine (e.g., dosage, side effects) once the medicine/pill is identified. These APIs do not use Core ML.

NOTE: The screen designs are made with help of "**draw.io**" tool. During the coding stage, there could be additional APIs learnt in class or Apple provided APIs used if needed depending on the state and requirements of the app.

References -

1. **Mobile First Design** -
   - https://www.geeksforgeeks.org/mobile-first-design/
   - https://www.uxpin.com/studio/blog/a-hands-on-guide-to-mobile-first-design/
2. **OpenFDA API** - https://open.fda.gov/apis/
3. **RxNorm API** - https://lhncbc.nlm.nih.gov/RxNav/APIs/RxNormAPIs.html